



MIPI D-PHY Bandwidth Matrix and Implementation

Technical Note

FPGA-TN-02090-1.2

June 2019

Disclaimers

Lattice makes no warranty, representation, or guarantee regarding the accuracy of information contained in this document or the suitability of its products for any particular purpose. All information herein is provided AS IS and with all faults, and all risk associated with such information is entirely with Buyer. Buyer shall not rely on any data and performance specifications or parameters provided herein. Products sold by Lattice have been subject to limited testing and it is the Buyer's responsibility to independently determine the suitability of any products and to test and verify the same. No Lattice products should be used in conjunction with mission- or safety-critical or any other application in which the failure of Lattice's product could create a situation where personal injury, death, severe property or environmental damage may occur. The information provided in this document is proprietary to Lattice Semiconductor, and Lattice reserves the right to make any changes to the information in this document or to any products at any time without notice.

Contents

Acronyms in This Document	5
1. Introduction	6
2. Video Format	7
2.1. Video Resolution and Pixel Clock	8
2.2. Color Depth	9
3. MIPI CSI-2/DSI Interfaces	10
4. Packetizing	14
4.1. xMulti-lane	15
5. Bandwidth and Data Rate	16
5.1. Bandwidth and Data Rate Calculation	16
5.1.1. Pixel Clock	16
5.1.2. Total Data Rate or Bandwidth	16
5.1.3. Data Rate per Lane	16
5.1.4. Bit Clock	16
5.2. Examples	16
5.2.1. Example 1: 1920x1080p@60Hz, RAW10, 2-lane	16
5.2.2. Example 2: 3840x2160@30Hz, RAW8, 4-lane	16
6. Device Selection	17
6.1. Hardware Features	17
7. MIPI Data Rate Calculation	19
7.1. FPGA Receiver Interface	19
7.1.1. MachXO2, MachXO3L/LF, and MachXO3D	19
7.1.2. LatticeECP3	20
7.1.3. ECP5/ECP5-5G	21
7.1.4. CrossLink Soft D-PHY	22
7.1.5. t_{SU}/t_{HD} Valid Window at Higher Data Rate	22
7.2. FPGA Transmitter Interface	23
7.2.1. MachXO2, MachXO3L/LF, and MachXO3D	23
7.2.2. LatticeECP3	24
7.2.3. ECP5/ECP5-5G	25
7.2.4. Tskew Window at Higher Data Rate	25
7.3. MIPI D-PHY Lane Number Selection Matrix Table	26
Reference	28
Technical Support	28
Revision History	28

Figures

Figure 1.1. CMOS Sensor Bridge Model	6
Figure 2.1. Interlaced Mode Video Frame Format	7
Figure 2.2. Most Common Display Resolutions	8
Figure 3.1. Unidirectional Receive HS Mode and Bidirectional LP Mode Interface Implementation	10
Figure 3.2. Unidirectional Receive HS Mode Only Implementation	11
Figure 3.3. Unidirectional Transmit HS Mode and Bidirectional LP Mode Interface Implementation	12
Figure 3.4. Unidirectional Transmit HS Mode Only Implementation	13
Figure 4.1. An Example of RAW10 Transmissions on CSI-2 Bus	14
Figure 7.1. MachXO2, MachXO3L/LF, and MachXO3D Maximum Data Rate	19
Figure 7.2. LatticeECP3 Maximum Data Rate	20
Figure 7.3. ECP5/ECP5-5G Maximum Data Rate	21
Figure 7.4. CrossLink Maximum Data Rate	22
Figure 7.5. MachXO2 Maximum Data Rate	23
Figure 7.6. LatticeECP3 Maximum Data Rate	24
Figure 7.7. ECP5/ECP5-5G Maximum Data Rate	25

Tables

Table 2.1. Common Video Format	9
Table 4.1. CSI-2 Packet Size Constraints*	15
Table 6.1. MIPI D-PHY RX/TX Implementation Hardware Comparison	17
Table 7.1. t_{SU}/t_{HD} Window for Higher Data Rate	23
Table 7.2. Tskew Window for Higher Data Rate	25
Table 7.3. MIPI D-PHY Interface Lane Number and Line Rate Selection Example Matrix Table	26

Acronyms in This Document

A list of acronyms used in this document.

Acronym	Definition
CSI	Camera Serial Interface
DE	Data Enable
DMT	Display Monitor Timing
DSI	Display Serial Interface
EAV	End of Active Video
FPGA	Field-Programmable Gate Array
HS	High Speed; Horizontal Sync
LP	Low Power
MIPI	Mobile Industry Processor Interface
SAV	Start of Active Video
VS	Vertical Sync

1. Introduction

As we move from the world of standard-definition to the high-definition and ultra-high-definition, the common parallel interfaces are difficult to expand, require many interconnects and consume relatively large amounts of power. Emerging packet-based serial interfaces, such as MIPI CSI-2 and DSI address many of the shortcomings of the parallel interfaces, while also introducing system complexity. Understanding the mathematics behind the parallel and serial interface bandwidth estimation can prevent a lot of problems when choosing an FPGA device with the right number of data lanes supporting the required data transfer rate. This document describes in details the methods of calculating the bandwidth and data rate of the image sensor's output of the RGB, YUV, or RAW data over a single or multi-lane MIPI CSI-2 and DSI interface. The same calculation method can be applied to other video interface such as FPD-Link, HiSPI, and HDMI.

Figure 1.1 shows a conceptual model of the CMOS Sensor Bridge Design. On the left, a CMOS sensor transfers image data to the FPGA through 1 to 4 serial data lanes; the FPGA sensor bridge merges the image data from multiple lanes and converts them into parallel data; on the right, the image data are sent out over the parallel bus in standard video format. Based on the known video format information, we can calculate the required bandwidth. Because the FPGA does not buffer the video frames, the peak transfer rate of CMOS sensor input must meet the bandwidth requirement of the output. With this presumption, we can estimate the maximum data rate and bit clock frequency of the CMOS sensor interface.

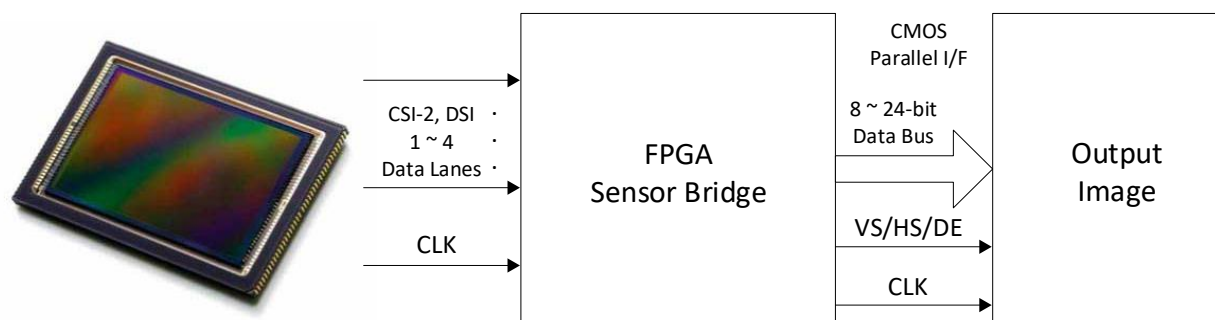


Figure 1.1. CMOS Sensor Bridge Model

2. Video Format

To estimate the data transfer rate, we need to understand the format of the video data transferred over the sensor bridge. Video is composed of a series of still images. Each still image is composed of individual lines of pixel data. [Figure 2.1](#) illustrates a conceptual interlaced video frame. The progressive video frame is similar to it except for only one field per frame.

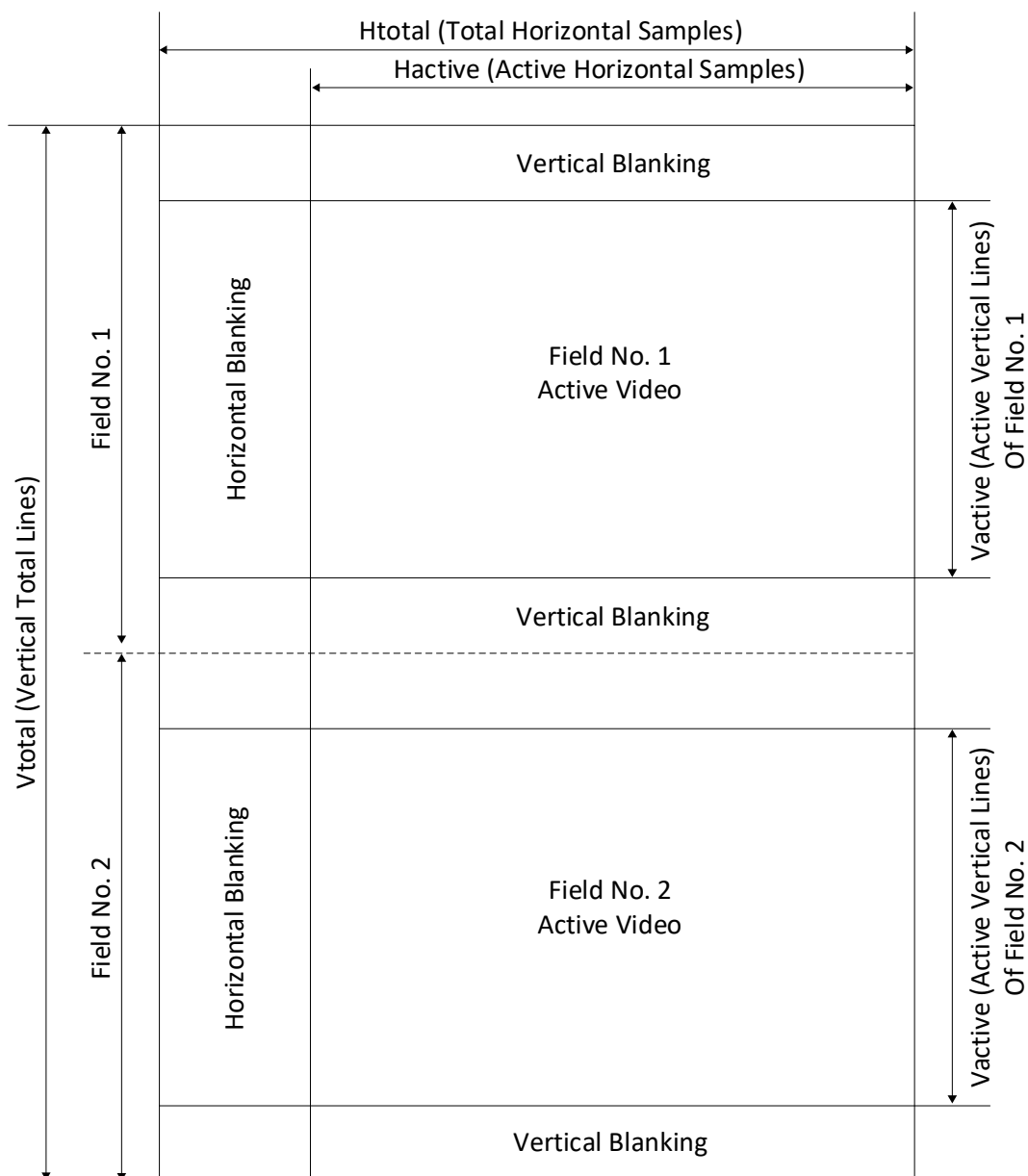


Figure 2.1. Interlaced Mode Video Frame Format

For digital video, either a separate Horizontal Sync (HS), Vertical Sync (VS) and Data Enable (DE) signals are used to synchronize the video data transfer, or a special sequence embedded into the video data stream indicating the Start of Active Video (SAV) or End of Active Video (EAV). For MIPI CSI-2, two packets structures are defined for Low Level Protocol layer: Long packets to carry payload data, and the Short Packets for Frame Synchronization (that is Frame Start and Frame End) and Line Synchronization (that is Line Start and Line End).

There are many characteristics of the video streams such as frame rate, interlaced vs progressive, aspect ratio, stereoscopic, color space, and color depth. We will review the major parameters that will be used in the calculation of the data transfer rate in the later sections of the document.

2.1. Video Resolution and Pixel Clock

The video resolution is quoted as Width x Height, with the unit in pixels: for example, 1920 x 1080 means the horizontal width is 1920 pixels and the vertical height is 1080 lines. Figure 2.2 shows the chart of the common display resolutions.

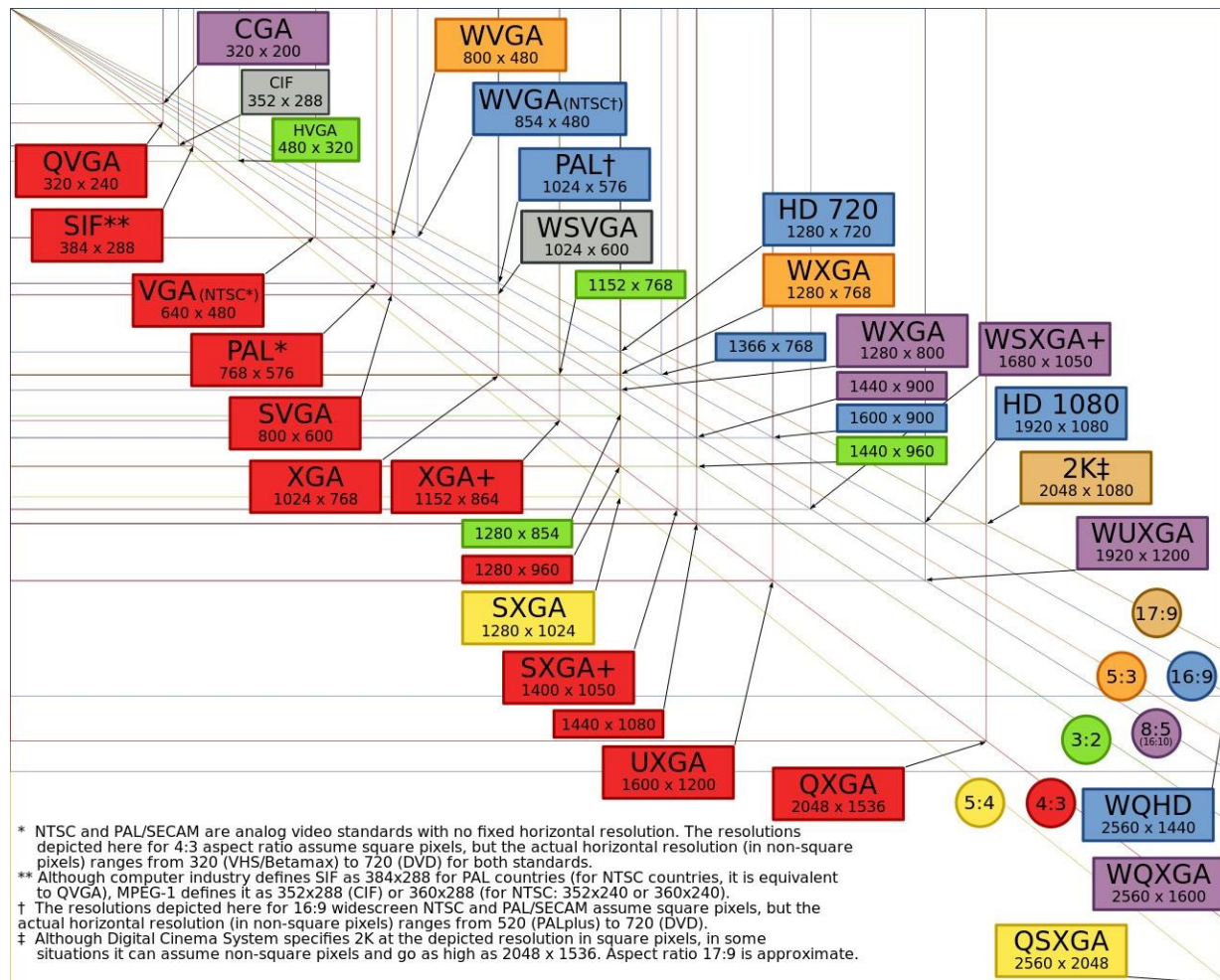


Figure 2.2. Most Common Display Resolutions

Source: https://en.wikipedia.org/wiki/File:Vector_Video_Standards4.svg

There are two major types of video format standards: SMPTE/CEA defines video standards for the Television and broadcast; VESA Display Monitor Timing (DMT) standard defines the video standards for the computer monitors. Table 2.1 lists the most common video resolutions. Among them, we choose HD (1280 x 720p), FHD (1920 x 1080p) and UHD (3840 x 2160p) as examples to calculate the bandwidth and data rate in the later sections.

Table 2.1. Common Video Format

		Hactive	Vactive	Htotal	Vtotal	Refresh Rate (Hz)	Pixel Freq (MHz)
Television Video Format							
EDTV	640x480p @ 59.94 Hz	640	480	800	525	59.94	25.175
	720x480p @ 59.94 Hz	720	480	858	525	59.94	27
	720x576p @ 50 Hz	720	576	864	625	50	27
HDTV	1280x720p @ 60 Hz	1280	720	1650	750	60	74.25
Full HD	1920x1080i @ 60 Hz	1920	1080	2200	1125	60	74.25
	1920x1080p @ 60 Hz	1920	1080	2200	1125	60	148.5
UHDTV	3840x2160p @ 60 Hz	3840	2160	4400	2250	60	594
	4096x2160p @ 60 Hz	4096	2160	4400	2250	60	594
Computer Monitor Format							
XGA	1024x768p @ 60 Hz	1024	768	1344	806	60	65
WXGA	1280x800p @ 60 Hz	1280	800	1680	831	59.81	83.5
WXGA+	1440x900p @ 60 Hz	1440	900	1904	934	59.887	106.5
HD	1366x768p @ 60 Hz	1366	768	1792	798	59.79	85.5
HD+	1600x900p @ 60 Hz (RB)	1600	900	1800	1000	60	108
WUXGA	1920x1200p @ 60 Hz (RB)	1920	1200	2080	1235	59.95	154
WQXGA	2560x1600p @ 60 Hz (RB)	2560	1600	2720	1646	59.97	268.5

Some of the image sensors can output the standard video formats by cropping or binning the pixels, while the others may output non-standard resolutions. The important thing is to obtain the information of the Total Horizontal Samples, Total Vertical Lines, and frame Refresh Rate. This document discusses how to calculate the bandwidth based on the above information.

2.2. Color Depth

Color depth, also known as bit depth, can be either referred to as bits-per-pixel (bpp) which specifies the number of bits used for a single pixel, or referred to as bits-per-component (bpc) which specifies the number of bits used to represent each color component of a single pixel. Deep color supports 30/36/48-bit for three RGB colors. In this document, the term Pixel Size is equivalent to the color depth in bits-per-pixel. For example, the pixel size of a 30-bit deep color RGB is defined as 30 bits per pixel, or 10 bits per color component.

3. MIPI CSI-2/DSI Interfaces

MIPI Camera Serial Interface 2 (CSI-2) and Display serial Interface (DSI) are two of the serial interface protocols based on MIPI D-PHY physical interface. MIPI D-PHY supports unidirectional HS (High Speed) mode and Bidirectional LP (Low Power) mode. For the application of CMOS sensor bridge, we only need the MIPI D-PHY receiver (RX) on the FPGA receive interface, which allows the bridge to receive HS data on one clock lane and up to four data lanes. Figure 3.1 shows the block diagram of the Unidirectional Receive HS Mode and Bidirectional LP Mode interface. Figure 3.2 shows the block diagram of the Unidirectional Receiver HS Mode Only interface. Figure 3.3 shows the block diagram of the Unidirectional Transmit HS Mode and Bidirectional LP Mode interface. Figure 3.4 shows the block diagram of the Unidirectional Transmit HS Mode Only interface.

Note: Figure 3.1 and Figure 3.2 do not apply to CrossLink since the CrossLink device uses dedicated MIPI D-PHY Input buffers for both HS and LP mode to meet MIPI input voltage specification. Figure 3.3 and Figure 3.4 also do not apply to CrossLink as CrossLink's programmable I/O can only be configured as a MIPI D-PHY receiver (soft MIPI D-PHY). Only the hardened MIPI D-PHY blocks can be configured as a transmitter.

For details on both Receiver/Transmitter HS and LP modes interface implementation, refer to Lattice reference design document [MIPI D-PHY Interface IP \(RD1182\)](#).

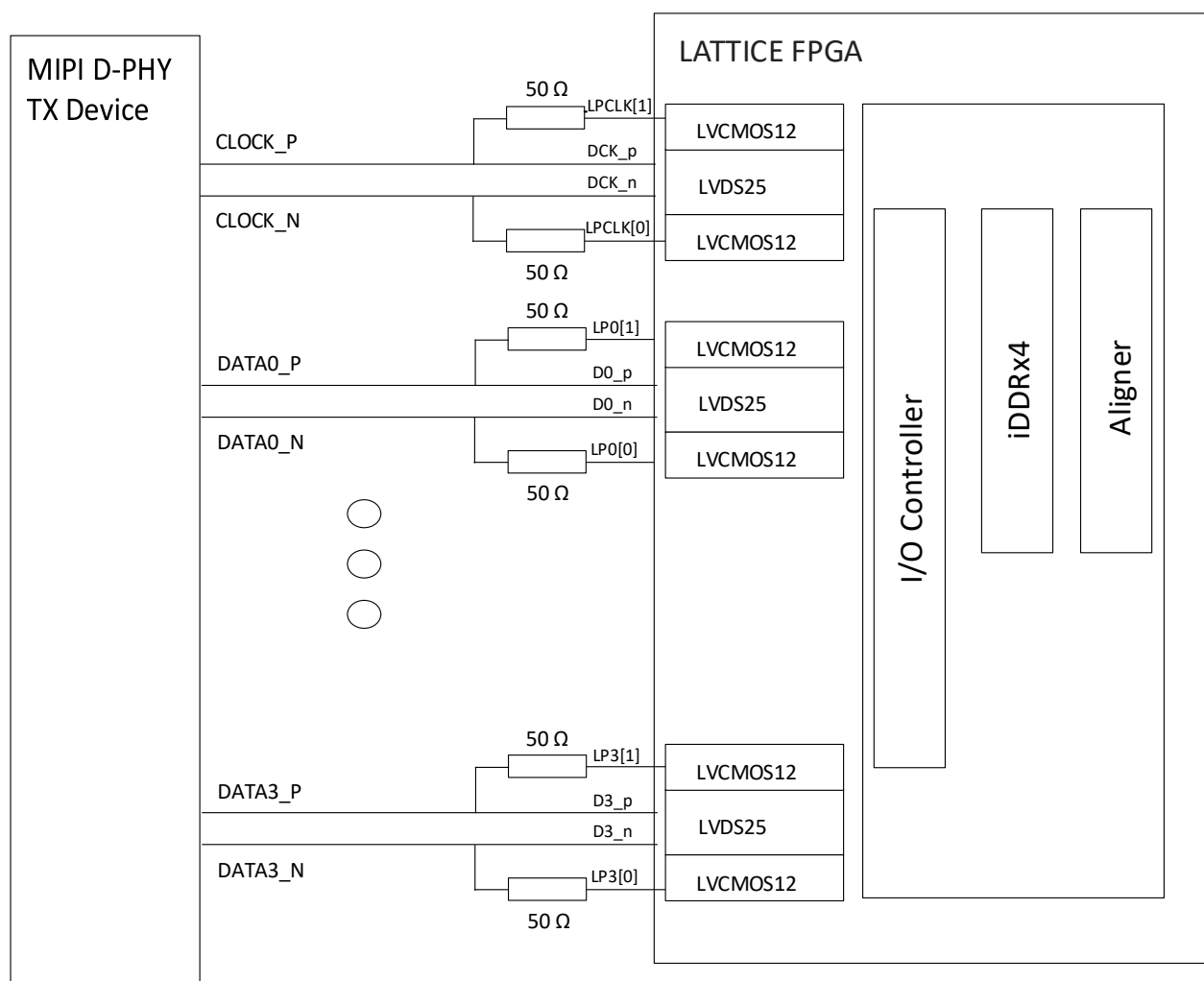


Figure 3.1. Unidirectional Receive HS Mode and Bidirectional LP Mode Interface Implementation

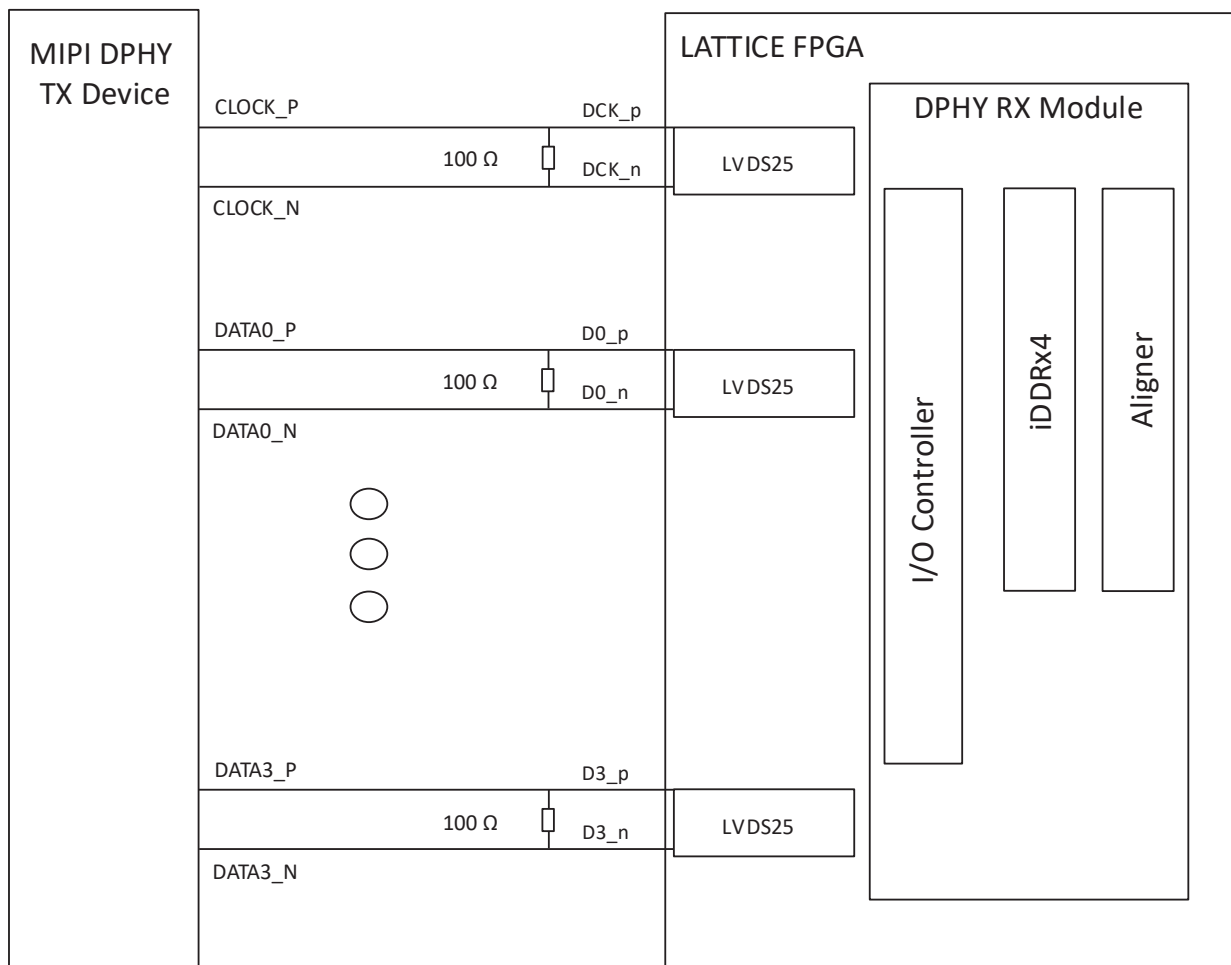


Figure 3.2. Unidirectional Receive HS Mode Only Implementation

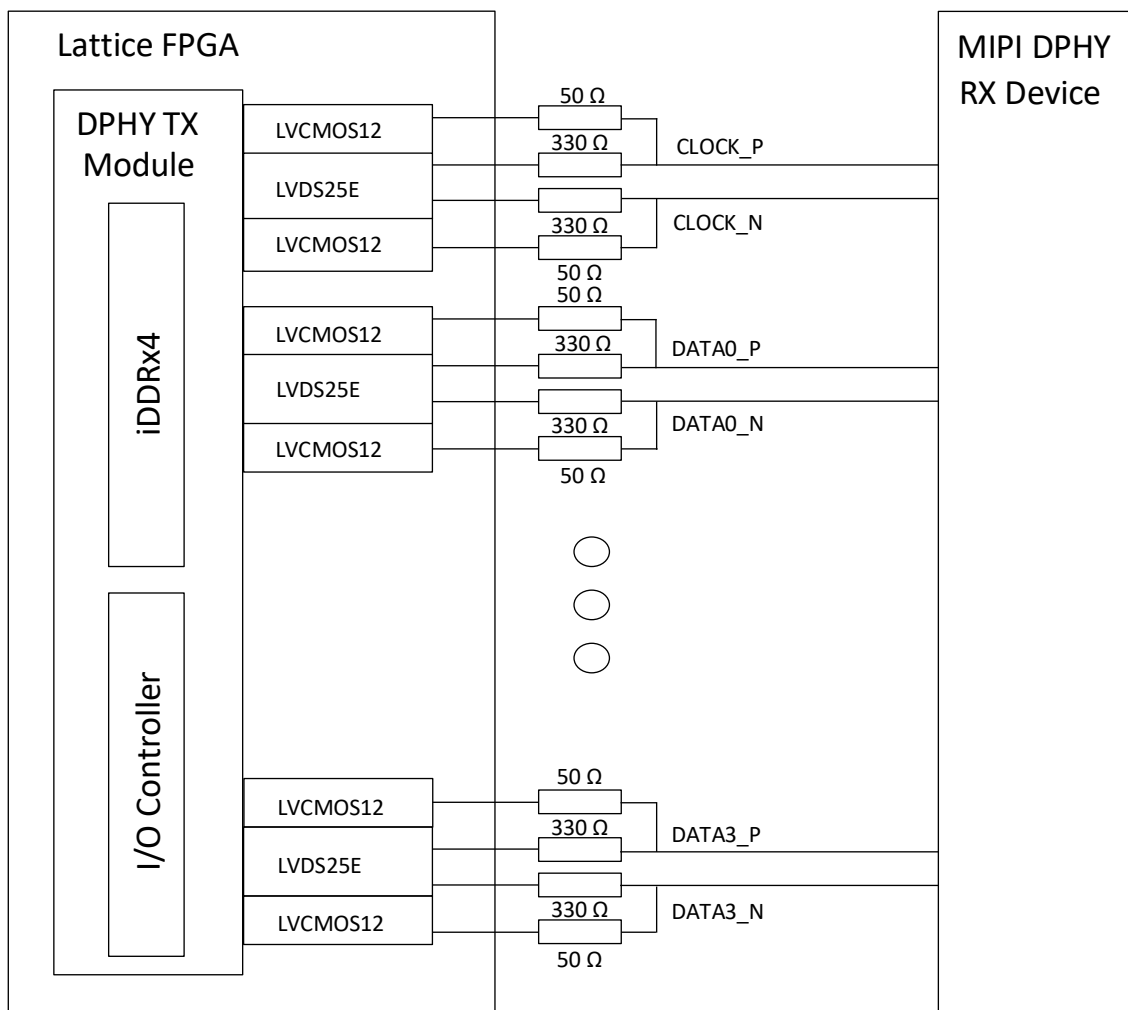


Figure 3.3. Unidirectional Transmit HS Mode and Bidirectional LP Mode Interface Implementation

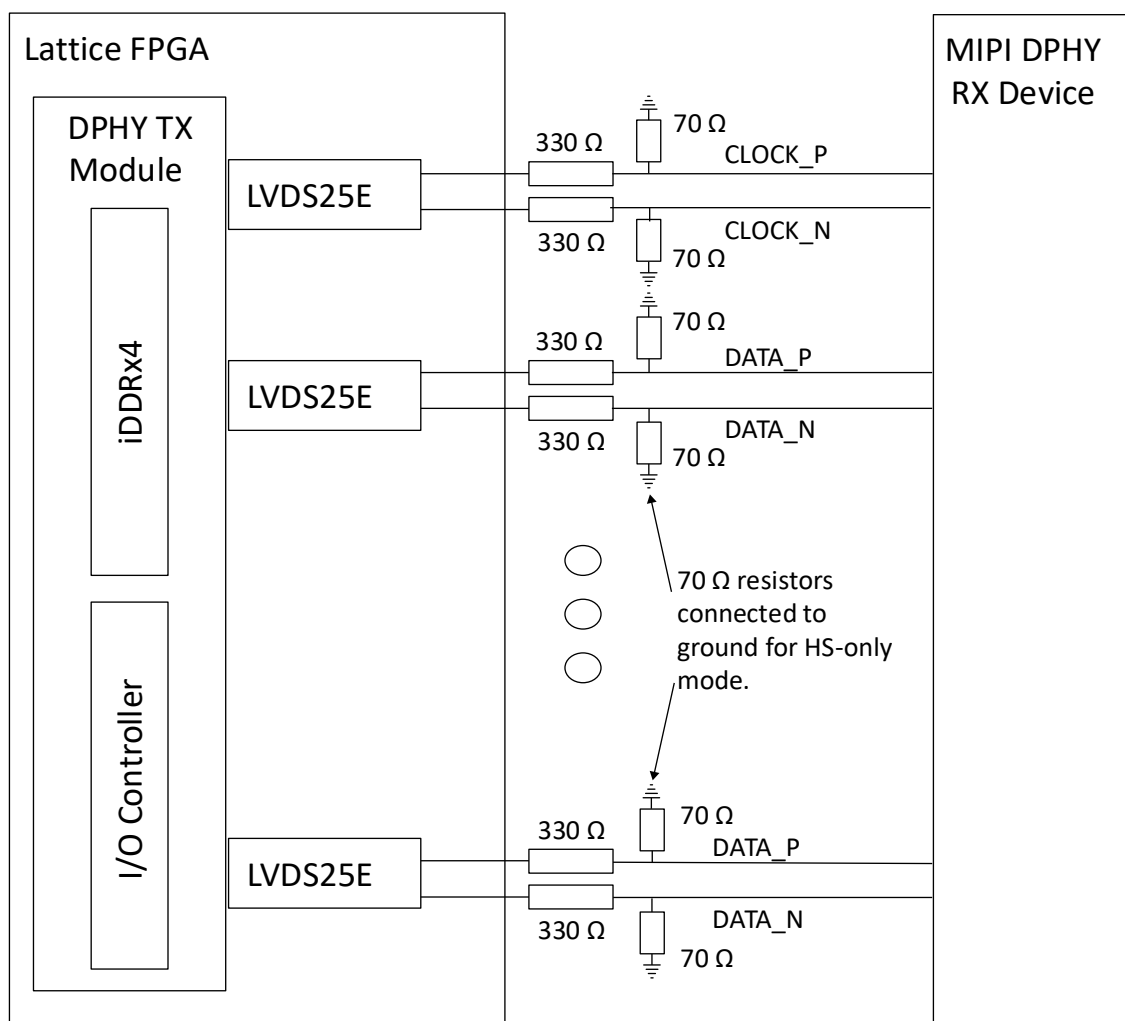


Figure 3.4. Unidirectional Transmit HS Mode Only Implementation

4. Packetizing

The MIPI CSI-2 supports YUV, RGB or RAW data with varying pixel formats from 6 to 24 bits per pixel. In the transmitter, the pixels are packed into bytes (Pixel-to-byte Packing) before sending the data to Low Level Protocol layer; in the receiver the bytes received from the Low Level Protocol layer are unpacked into pixel (Byte-to-pixel Unpacking). One CSI-2 long packet shall contain one line of image data. The total size of data within a long packet for all data types shall be a multiple of eight bits (byte). [Figure 4.1](#) shows an example of packing four 10-bit pixel data, or RAW10, into five bytes to look like 8-bit data format.

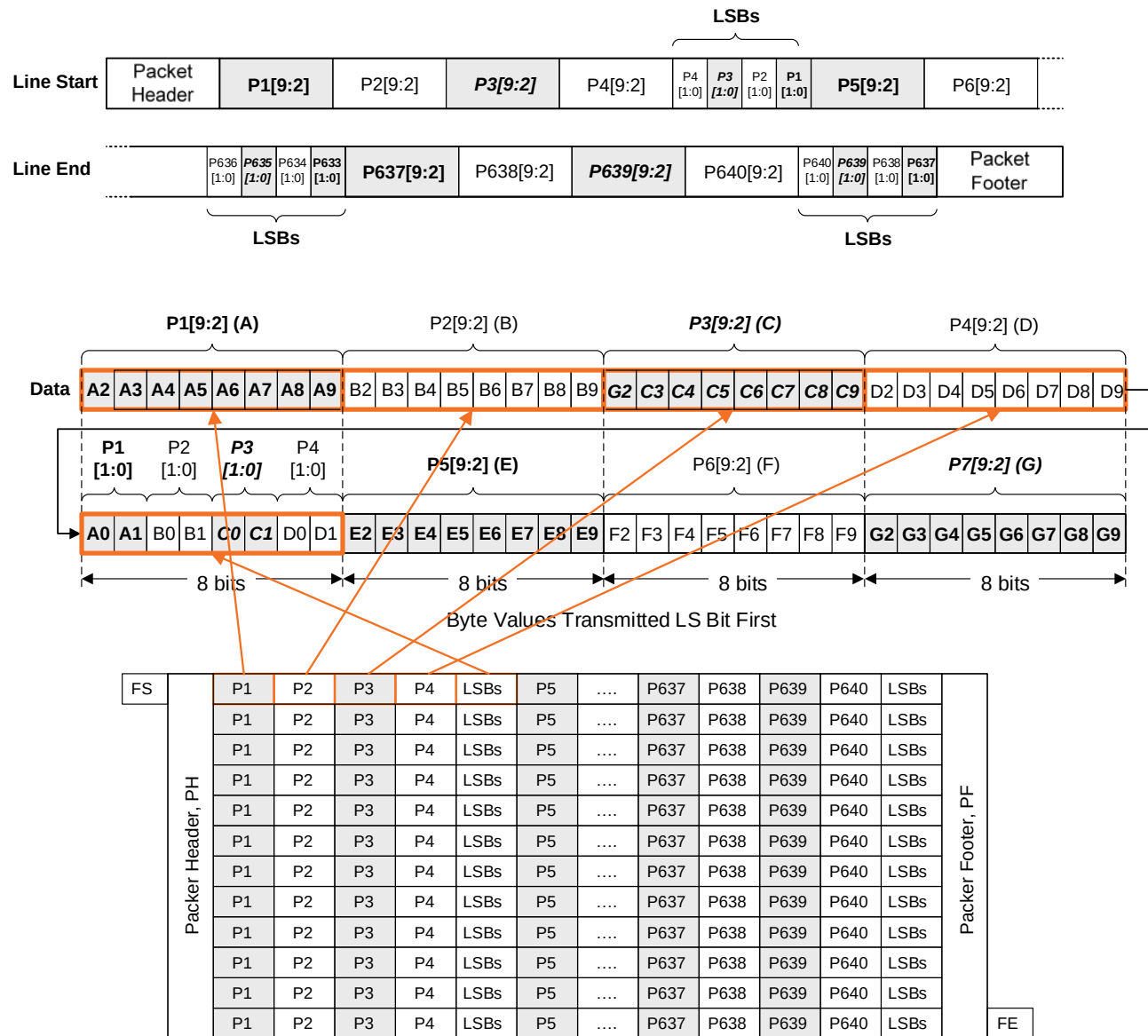


Figure 4.1. An Example of RAW10 Transmissions on CSI-2 Bus

Table 4.1 specifies the packet size constraints for the supported data formats. The length of each packet must be a multiple of the values in the table. For simplicity, all data format examples are single lane configurations.

Table 4.1. CSI-2 Packet Size Constraints*

Data Format	Odd/Even Lines	Pixels	Bytes	Bits
YUV420 8-bit	Odd	2	2	16
	Even	2	4	32
YUV420 10-bit	Odd	4	5	40
	Even	4	10	80
YUV422 8-bit	—	2	4	32
YUV422 10-bit	—	2	5	40
RGB888	—	1	3	24
RGB666	—	4	9	72
RGB565	—	1	2	16
RGB555	—	1	2	16
RGB444	—	1	2	16
RAW6	—	4	3	24
RAW7	—	8	7	56
RAW8	—	1	1	8
RAW10	—	4	5	40
RAW12	—	2	3	24
RAW14	—	4	7	56
RAW16	—	1	2	16
RAW20	—	2	5	40

***Note:** RGB555 and RGB444 data should be made to look like RGB565 data by inserting padding bits to the LSBs of each color component.

4.1. xMulti-lane

MIPI CSI-2 is lane-scalable. Applications requiring more bandwidth than that provided by one data lane, or those trying to avoid high clock rates, can expand the data path to two, three, or four lanes wide and obtain approximately linear increases in peak bus bandwidth

5. Bandwidth and Data Rate

This section provides the definitions of terms used in video performance and summarizes the process of calculating bandwidth and data transfer rate.

- **Pixel Clock** – the time base in MHz at which individual pixels are transmitted
- **Bandwidth** – the capacity required in Mbps of a given system to pass a specific frequency
- **Data Rate** – the data flow throughput in bits per second of the transport layer

5.1. Bandwidth and Data Rate Calculation

5.1.1. Pixel Clock

If the video format is standard, the pixel clock frequency can be obtained from the SMPTE/CEA or VESA standards. You can also calculate the pixel clock frequency using the following equation:

$$\text{Pixel Clock Frequency} = \text{Total Horizontal Samples} * \text{Total Vertical Lines} * \text{Refresh Rate}$$

The Total Horizontal Samples and Total Vertical Lines include blanking period. The Refresh Rate may be referred to as Frame Rate or Vertical Frequency

5.1.2. Total Data Rate or Bandwidth

The bandwidth of a given video format is simply a product of the Pixel Clock Frequency and Pixel Size in bits. The total data rate of the CMOS sensor interface must match the bandwidth.

$$\text{Total Data Rate (Bandwidth)} = \text{Pixel Clock Frequency} * \text{Pixel Size (in bits)}$$

5.1.3. Data Rate per Lane

The per-lane data rate (Line Rate) is the total data rate (bandwidth) divided by the number of lanes. CSI-2 can support up to four data lanes.

$$\text{Data Rate per Lane} = \text{Total Data Rate (Bandwidth)} / \text{Number of Data Lane}$$

5.1.4. Bit Clock

Because the MIPI data lane is a Double Data Rate interface, the CSI-2 Bit Clock frequency is one half of the Data Rate per Lane.

$$\text{Bit Clock Frequency} = \text{Data Rate per Lane} / 2$$

5.2. Examples

5.2.1. Example 1: 1920x1080p@60Hz, RAW10, 2-lane

Total Horizontal Samples = 2200, Total Vertical Lines = 1125

Pixel Clock Frequency = $2200 \times 1125 \times 60 = 148.5$ MHz

Bandwidth (Total Data Rate) = $148.5 \text{ MHz} * 10\text{-bit} = 1.485$ Gbps

Line Rate (Data Rate per Lane) = $1.485 \text{ Gbps} / 2\text{-lane} = 742.5$ Mbps

MIPI Bit Clock Frequency = $742.5 / 2 = 371.25$ MHz

5.2.2. Example 2: 3840x2160@30Hz, RAW8, 4-lane

Total Horizontal Samples = 4400, Total Vertical Lines = 2250

Pixel Clock Frequency = $4400 \times 2250 * 30 = 297$ MHz

Bandwidth (Total Data Rate) = $297 \text{ MHz} * 8\text{-bit} = 2.376$ Gbps

Line Rate (Data Rate per Lane) = $2.376 \text{ Gbps} / 4\text{-lane} = 594$ Mbps

MIPI Bit Block Frequency = $594 / 2 = 297$ MHz

6. Device Selection

This section summarizes Lattice FPGA families that support MIPI interface with external components as described in the [MIPI CSI-2/DSI Interfaces](#) section.

6.1. Hardware Features

[Table 6.1](#) highlights the features of MIPI D-PHY hardware implementation using MachXO2, MachXO3L/LF, MachXO3D, LatticeECP3, ECP5/ECP5-5G and Crosslink device families. For more details on MIPI D-PHY Receiver and Transmitter resource utilization and design performance, refer to the [MIPI D-PHY Interface IP \(RD1182\)](#) document.

Table 6.1. MIPI D-PHY RX/TX Implementation Hardware Comparison

		MachXO2, MachXO3L/LF, and MachXO3D (Soft)	LatticeECP3 (Soft)	ECP5/ECP5-5G (Soft)	CrossLink ³ (Soft)	CrossLink ³ (Hard)
MIPI D-PHY Rx Implementations	HS Mode	- LVDS25 - VCCIO = 2.5 V or 3.3 V - Internal 100 Ω differential termination.	- LVDS25 - VCCIO = 2.5 V or 3.3 V - Internal 100 Ω differential termination.	- LVDS25 - VCCIO = 2.5 V or 3.3 V - Internal 100 Ω differential termination. - Or use IMIPI primitive (C/D pair)	- Use MIPI_DPHY primitive - Internal 100 Ω differential termination.	—
	LP Mode	- LVC MOS12 - VCCIO = 1.2 V	- LVC MOS12 - VCCIO = 1.2 V	- LVC MOS12 - VCCIO = 1.2 V - Or use IMIPI primitive (C/D pair)	Use MIPI_DPHY primitive	—
MIPI D-PHY Tx Implementations	HS Mode	LVDS25E	LVDS25E or LVC MOS33D²	LVDS25E or LVC MOS33D²	—	—
	LP Mode	- LVC MOS12 - VCCIO = 1.2 V	- LVC MOS12 - VCCIO = 1.2 V	- LVC MOS12 - VCCIO = 1.2 V	—	—
DDR Gearing Ratio		X4	X2	X2	X4 or X8	X4 or X8
Number of D-PHY	Rx	Up to two RX D-PHYs; Bank 2 only.	Up to two RX D-PHYs. Only two clock divider primitives, one on Bank 2 or Bank 3 (right), one on Bank 6 or Bank 7 (left)	Up to four RX D-PHYs (left and right sides); Four dedicated Clock Divider (CLKDIV) available.	Limited by PIO pairs and clock divider available in Bank 1 and Bank 2 only	—
	Tx	TX D-PHY max limited by PIO A/B pairs; Bank 0 only	TX D-PHY maximum limited by PIO pairs available in Bank 2, Bank 3, Bank 6, and Bank 7 if CLKDIV is shared on each side. If TX D-PHY on side, can only have RX D-PHY on the other.	TX D-PHY maximum limited by PIO pairs, available in the left and right banks.	—	—
Number of Lanes		Depends on the number of PIO A/B pairs available.	Depends on the number of PIO pairs available.	Depends on the number of PIO pairs available.	Depends on the number of PIO pairs available.	4 lanes max
RX Performance ¹		523 Mbps	467 Mbps	467 Mbps	1.2 Gbps	1.5 Gbps
TX Performance ¹		728 Mbps	698 Mbps	800 Mbps	—	1.5 Gbps

Notes:

1. Rx Performance and Tx Performance values are calculated from the data sheet External Timing Table using IDDRX2/X4/X8 and ODDRX2/X4/X8 with data and clock centered aligned. The maximum data rate each family device can achieve is compliant with MIPI Alliance Specification (v1.1). This is ± 0.15 UI setup/hold window on the receiver at ≤ 1 Gbps and ± 0.15 UI skew window on the transmitter at ≤ 1 Gbps. See the [MIPI Data Rate Calculation](#) section for the calculations based on current data sheet. It is still practical to reach performance higher than the specified data rate. Refer to the [tSU/tHD Valid Window at Higher Data Rate](#) section and the [Tskew Window at Higher Data Rate](#) section for details.
2. If LVCMOS33D is used instead of LVDS25E, external resistors shown in [Figure 3.3](#) and [Figure 3.4](#) should be adjusted to modulate common mode voltage level.
3. CrossLink has two hardened MIPI IPs on the top side of the chip, and has dedicated I/O associated with hardened IP. With regards to MIPI D-PHY interface implementation, on the receiver side, it has hardened IP offering up to 1.5 Gbps as well as soft IP offering up to 1.2 Gbps. On the transmitter side, however it only offers hardened IP implementation up to 1.5 Gbps. Refer to [CrossLink High-Speed I/O MIPI D-PHY and DDR interfaces \(FPGA-TN-02012\)](#) for details.

7. MIPI Data Rate Calculation

This section shows the calculations of the maximum data rate that can be achieved on Lattice FPGA products. The values are based on data sheet specification of DDRX2/X4/X8 with clock and data center-aligned. These are MIPI Alliance Specification (v1.1) compliant with ± 0.15 UI setup/hold window on the receiver at ≤ 1 Gbps and ± 0.15 UI skew window on the transmitter at ≤ 1 Gbps.

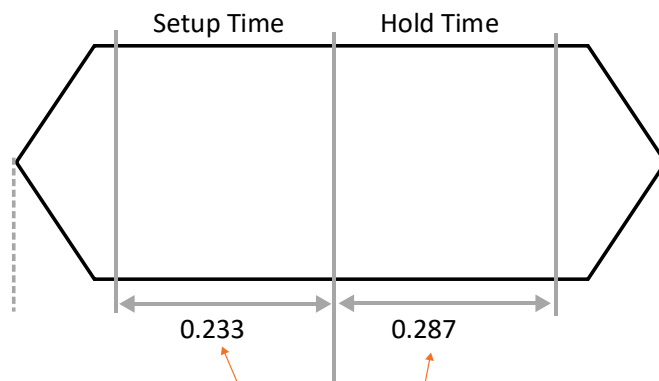
This section also provides the method of calculating the window if higher data rate is desired. The window for higher data rate does not meet MIPI Alliance Specification, but can still be practical in the user's implementation. Check the latest data sheet on the maximum data rate each Lattice FPGA device family can achieve.

7.1. FPGA Receiver Interface

7.1.1. MachXO2, MachXO3L/LF, and MachXO3D

Maximum MIPI compliance data rate calculation for MachXO2, MachXO3L/LF, and MachXO3D:

- $\text{Max}(0.233, 0.287) = 0.15 \times \text{UI}$
- $\text{UI} = 0.287/0.15 = 1.913 \text{ ns}$
- $\text{Max Data Rate} = 1/\text{UI} = 1/1.913 = 523 \text{ Mbps (at } 0.15 \text{ UI)}$



MachXO2 Data Sheet Rev. 3.3 at 756 Mbps

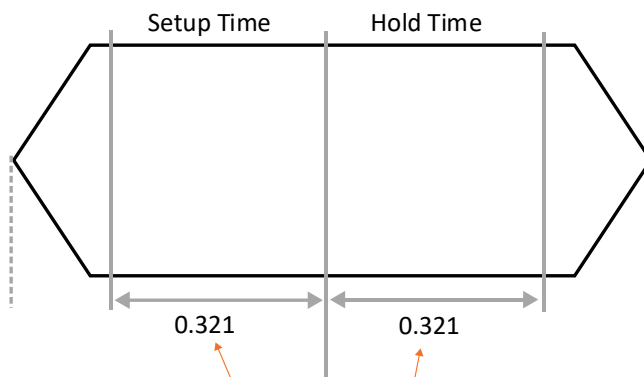
Generic DDRX4 Inputs with Clock and Data Centered at Pin Using PCLK Pin for Clock Input – GDDR4_RX.ECLK.Centered									
t_{SU}	Input Data Setup Before CLK	MachXO2-640U, MachXO2-1200/U and larger devices, bottom side only	0.233	—	0.219	—	0.198	—	ns
t_{HO}	Input Data Hold After CLK		0.287	—	0.287	—	0.344	—	ns
f_{DATA}	DDR4 Serial Input Data Speed		—	756	—	630	—	524	Mbps
f_{DDR4}	DDR2 ECLK Frequency		—	378	—	315	—	262	MHz
f_{SCLK}	SCLK Frequency		—	95	—	79	—	66	MHz

Figure 7.1. MachXO2, MachXO3L/LF, and MachXO3D Maximum Data Rate

7.1.2. LatticeECP3

Maximum MIPI compliance data rate calculation for LatticeECP3:

- $\text{Max}(0.321, 0.321) = 0.15 \times \text{UI}$
- $\text{UI} = 0.321/0.15 = 2.14 \text{ ns}$
- $\text{Max Data Rate} = 1/\text{UI} = 1/2.14 = 467 \text{ Mbps (at } 0.15 \text{ UI)}$



LatticeECP3 Data Sheet Rev. 2.8EA at 800 Mbps

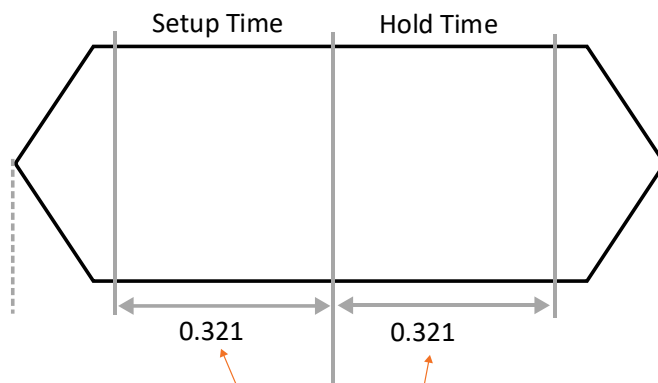
Generic DDRX2 Inputs with Clock and Data (>10 Bits Wide) Centered at Pin (GDDR2_RX.CLK.Centered) Using PCLK Pin for Clock Input									
Left and Right Sides									
t_{SUGDDR}	Data Setup Before CLK	ECP3-150EA	321	—	403	—	471	—	ps
t_{HOGDDR}	Data Hold After CLK	ECP3-150EA	321	—	403	—	471	—	ps
$f_{\text{MAX_GDDR}}$	DDR2 Clock Frequency	ECP3-150EA	—	405	—	325	—	280	MHz
t_{SUGDDR}	Data Setup Before CLK	ECP3-70EA/95EA	321	—	403	—	535	—	ps
t_{HOGDDR}	Data Hold After CLK	ECP3-70EA/95EA	321	—	403	—	535	—	ps
$f_{\text{MAX_GDDR}}$	DDR2 Clock Frequency	ECP3-70EA/95EA	—	405	—	325	—	250	MHz
t_{SUGDDR}	Data Setup Before CLK	ECP3-35EA	335	—	425	—	535	—	ps
t_{HOGDDR}	Data Hold After CLK	ECP3-35EA	335	—	425	—	535	—	ps
$f_{\text{MAX_DDR}}$	DDR2 Clock Frequency	ECP3-35EA	—	405	—	325	—	250	MHz
t_{SUGDDR}	Data Setup Before CLK	ECP3-17EA	335	—	425	—	535	—	ps
t_{HOGDDR}	Data Hold After CLK	ECP3-17EA	335	—	425	—	535	—	ps
$f_{\text{MAX_GDDR}}$	DDR2 Clock Frequency	ECP3-17EA	—	405	—	325	—	250	MHz

Figure 7.2. LatticeECP3 Maximum Data Rate

7.1.3. ECP5/ECP5-5G

Maximum MIPI compliance data rate calculation for ECP5/ECP5-5G:

- $\text{Max}(0.321, 0.321) = 0.15 \times \text{UI}$
- $\text{UI} = 0.321/0.15 = 2.14 \text{ ns}$
- $\text{Max Data Rate} = 1/\text{UI} = 1/2.14 = 467 \text{ Mbps (at } 0.15 \text{ UI)}$



ECP5/ECP5-5G Data Sheet Rev. 1.9 at 800 Mbps

Generic DDRX2 Inputs with Clock and Data Centered at Pin (GDDR2_RX.ECLK.Centered) Using PCLK Clock Input, Left and Right Sides Only									
$t_{\text{SU_GDDR2_CENTERED}}$	Data Setup before CLK Input	All Devices	0.321	—	0.403	—	0.471	—	ns
$t_{\text{HD_GDDR2_CENTERED}}$	Data Hold after CLK Input	All Devices	0.321	—	0.403	—	0.471	—	ns
$f_{\text{DATA_GDDR2_CENTERED}}$	GDDR2 Data Rate	All Devices	—	800	—	700	—	624	Mb/s
$f_{\text{MAX_GDDR2_CENTERED}}$	GDDR2 CLK Frequency (ECLK)	All Devices	—	400	—	350	—	312	MHz

Figure 7.3. ECP5/ECP5-5G Maximum Data Rate

7.1.4. CrossLink Soft D-PHY

Maximum MIPI compliance data rate calculation for CrossLink:

- 1200 Mbps at $V_{IDTH} = 140$ mV, $V_{IDTL} = -140$ mV
- 900 Mbps at $V_{IDTH} = 140$ mV, $V_{IDTL} = -140$ mV
- 600 Mbps at $V_{IDTH} = 70$ mV, $V_{IDTL} = -70$ mV

CrossLink Data Sheet Rev. 1.5

General Purpose I/O MIPI D-PHY Rx with 1:8 or 1:16 Gearing					
$t_{SU_GDDR_MP}$	Input Data Set-Up Before CLK	900 Mb/s < Data Rate $\leq$ 1.2 Gb/s & $V_{IDTH} = 140$ mV $V_{IDTL} = -140$ mV	0.200	—	UI
		600 Mb/s < Data Rate $\leq$ 900 Mb/s & $V_{IDTH} = 140$ mV $V_{IDTL} = -140$ mV	0.150	—	UI
		Data Rate $\leq$ 600 Mb/s & $V_{IDTH} = 70$ mV $V_{IDTL} = -70$ mV	0.150	—	UI
$t_{HD_GDDR_MP}$	Input Data Hold After CLK	900 Mb/s < Data Rate $\leq$ 1.2 Gb/s & $V_{IDTH} = 140$ mV $V_{IDTL} = -140$ mV	0.200	—	UI
		600 Mb/s < Data Rate $\leq$ 900 Mb/s & $V_{IDTH} = 140$ mV $V_{IDTL} = -140$ mV	0.150	—	UI
		Data Rate $\leq$ 600 Mb/s & $V_{IDTH} = 70$ mV $V_{IDTL} = -70$ mV	0.150	—	UI
$f_{MAX_GDDR_MP}$	Frequency for ECLK ³	csfBGA81, ctfBGA80, ckfBGA80, ucfBGA64	—	600	MHz
		WLCSP36	—	500	MHz

Figure 7.4. CrossLink Maximum Data Rate

7.1.5. t_{SU}/t_{HD} Valid Window at Higher Data Rate

For data rate higher than specified above, the t_{SU}/t_{HD} window may exceed the 0.15 UI specified in the MIPI Alliance Specification.

To calculate the window, the following equation can be used:

$$t_{SU}/t_{HD} \text{ Window} = \max(t_{SU}/t_{HD}) / (1 / \text{Data Rate})$$

For 800 Mbps, $1 / \text{Data Rate} = 1.25$ ns.

The maximum data rate each device can support on the receiver is limited to the data rate supported with IDDRX2/X4/X8 with center-aligned data. This is summarized in Table 7.1.

Table 7.1. t_{SU}/t_{HD} Window for Higher Data Rate

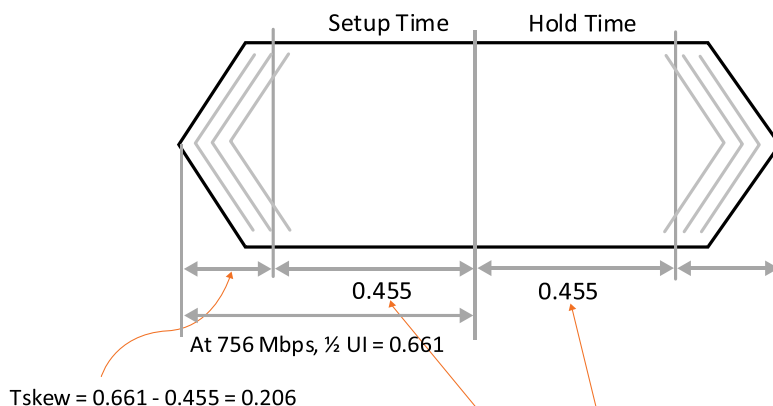
Device Family	Max Data Rate	Tsu/Thd Window
MachXO2	756 Mbps	0.217 UI
MachXO3L/LF and MachXO3D	900 Mbps	0.285 UI
LatticeECP3	800 Mbps	0.257 UI
ECP5	800 Mbps	0.257 UI
ECP5-5G	800 Mbps	0.257 UI
CrossLink Soft D-PHY	1.2 Gbps at $V_{IDTH} = 140$ mv, $V_{IDTL} = -140$ mv	0.20 UI

7.2. FPGA Transmitter Interface

7.2.1. MachXO2, MachXO3L/LF, and MachXO3D

Maximum MIPI compliance data rate calculation for MachXO2, MachXO3L/LF, and MachXO3D:

- $0.206 = 0.15 \times UI$
- $UI = 0.206/0.15 = 1.373$ ns
- Max Data Rate = $1/UI = 1/1.373 = 728$ Mbps (at 0.15 UI)



MachXO2 Data Sheet Rev. 3.3 at 756 Mbps

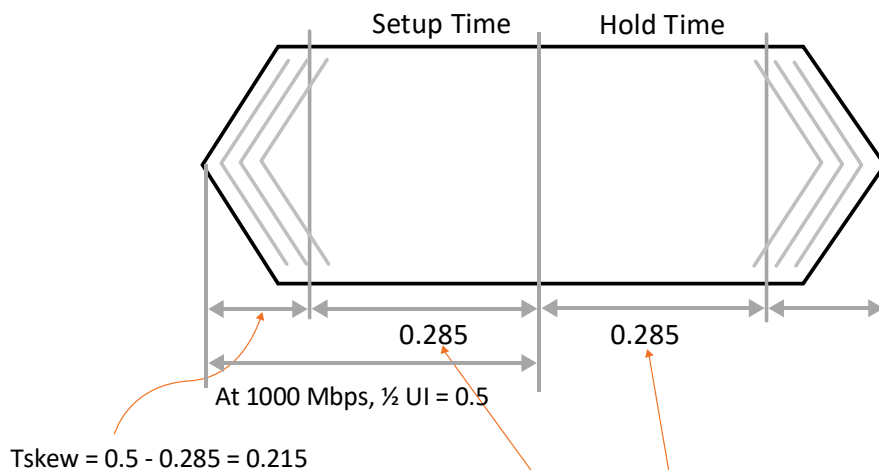
Generic D DRX4 Outputs with Clock and Data Centered at Pin Using PCLK Pin for Clock Input – GDDR4TX.ECLK.Centered									
t_{DVB}	Output Data Valid Before CLK Output	MachXO2-640U, MachXO2-1200/U and larger devices, top side only	0.455	—	0.570	—	0.710	—	ns
t_{DVA}	Output Data Hold After CLK Output		0.455	—	0.570	—	0.710	—	ns
f_{DATA}	DDR4 Serial Output Data Speed		—	756	—	630	—	524	Mbps
f_{DDR4}	DDR4 ECLK Frequency (minimum limited by PLL)		—	378	—	315	—	262	MHz
f_{CLK}	SCLK Frequency		—	95	—	79	—	66	MHz

Figure 7.5. MachXO2 Maximum Data Rate

7.2.2. LatticeECP3

Maximum MIPI compliance data rate calculation for LatticeECP3:

- $0.215 = 0.15 \times UI$
- $UI = 0.215/0.15 = 1.433 \text{ ns}$
- Max Data Rate = $1/UI = 1/1.433 = 698 \text{ Mbps (at } 0.15 \text{ UI)}$



LatticeECP3 Data Sheet Rev. 2.8EA at 1000 Mbps

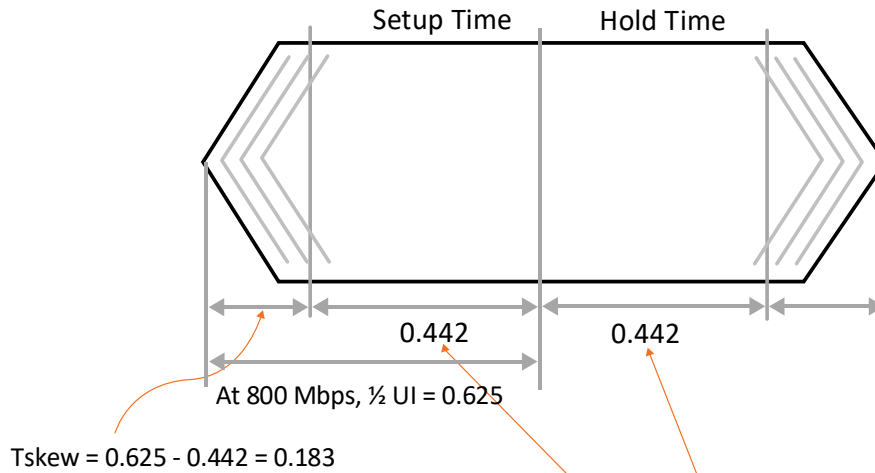
Generic DDRX2 Output with Clock and Data (>10 Bits Wide) Centered at Pin Using PLL (GDDR2_TX.PLL.Centered)									
Left and Right Sides									
$t_{DVBGDDR}$	Data Valid Before CLK	All ECP3EA Devices	285	—	370	—	431	—	ps
$f_{DVAGDDR}$	Data Valid After CLK	All ECP3EA Devices	285	—	370	—	432	—	ps
f_{MAX_GDDR}	DDR2 Clock Frequency	All ECP3EA Devices	—	500	—	420	—	375	MHz

Figure 7.6. LatticeECP3 Maximum Data Rate

7.2.3. ECP5/ECP5-5G

Maximum MIPI compliance data rate calculation for ECP5/ECP5-5G:

- $0.183 = 0.15 \times UI$
- $UI = 0.183/0.15 = 1.22 \text{ ns}$
- Max Data Rate = $1/UI = 1/1.22 = \sim 800 \text{ Mbps}$ (at 0.15 UI)



ECP5/ECP5-5G Data Sheet Rev. 1.9 at 800 Mbps

Generic DDRX2 Outputs with Clock and Data Centered at Pin (GDDR2_TX.ECLK.Centered) Using PCLK Clock Input, Left and Right Sides Only									
$t_{VB_GDDR2_centered}$	Data Output Valid Before CLK Output	All Devices	0.442	—	0.56	—	0.676	—	ns + $\frac{1}{2} UI$
$t_{VA_GDDR2_centered}$	Data Output Valid After CLK Output	All Devices	—	0.442	—	0.56	—	0.676	ns + $\frac{1}{2} UI$
$f_{DATA_GDDR2_centered}$	GDDR2 Data Rate	All Devices	—	800	—	700	—	624	Mb/s
$f_{MAX_GDDR2_centered}$	GDDR2 CLK Frequency (ECLK)	All Devices	—	400	—	350	—	312	MHz

Figure 7.7. ECP5/ECP5-5G Maximum Data Rate

7.2.4. Tskew Window at Higher Data Rate

For data rate higher than specified above, the skew window may exceed the 0.15 UI specified in the MIPI Alliance Specification.

To calculate the window, the following equation can be used:

$$\frac{1}{2} UI = \frac{1}{2} * (1 / \text{Data Rate})$$

$$\text{Tskew Window} = (\frac{1}{2} UI - \min(tDVA, tDVB))/UI$$

For 800 Mbps, $1 / \text{Data Rate} = 1.25 \text{ ns}$.

The maximum data rate each device can support on the transmitter is limited to the data rate supported with ODDR2/X4/X8 with center-aligned data. This can be summarized in Table 7.2.

Table 7.2. Tskew Window for Higher Data Rate

Device Family	Max Data Rate	Tsu/Thd Window
MachXO2	756 Mbps	0.155 UI
MachXO3L/LF and MachXO3D	900 Mbps	0.185 UI
LatticeECP3	800 Mbps	0.172 UI
ECP5	800 Mbps	$\leq 0.150 UI$
ECP5-5G	800 Mbps	$\leq 0.150 UI$

7.3. MIPI D-PHY Lane Number Selection Matrix Table

Table 7.3 lists the selection of MIPI D-PHY Lane number and Line rate versus the Total Data Rate required by each video format. For clarity, this Matrix table only chooses three common resolutions and the major color depths as examples. The lane number and line rate for other video formats can be calculated using the equations listed in Bandwidth and Data Rate section.

Table 7.3. MIPI D-PHY Interface Lane Number and Line Rate Selection Example Matrix Table

					MachXO2, MachXO3L/LF, and MachXO3D (523 Mbps for RX and 728 Mbps for TX I/O)			ECP3 (467 Mbps for RX and 698 Mbps for TX I/O)			ECP5/ECP5-5G (467 Mbps for RX and 800 Mbps for TX I/O)			CrossLink (1.5 Gbps for RX and 1.5 Gbps for TX I/O)		
Resolution	Frame Rate	Pixel Clock	Color Depth	Total Data Rate	Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock
	Hz	MHz	Bits	Mbps		Mbps	MHz		Mbps	MHz		Mbps	MHz		Mbps	MHz
HD 1280x720p (1650x750)	60	74.25	8	594	2	297	148.5	2	297	148.5	2	297	148.5	1	594	297
			10	742.5	2	371	185.6	2	371	185.6	2	371	185.6	1	742.5	371
			16	1188	4	297	148.5	4	297	148.5	4	297	148.5	2	594	297
			18	1336.5	4	334	167	4	334	167	4	334	167	2	668	334
			24	1782	4	445.5	223	4	445.5	223	4	445.5	223	2	891	445.5
	120	148.5	8	1188	4	297	148.5	4	297	148.5	4	297	148.5	2	594	297
			10	1485	4	371	185.6	4	371	185.6	4	371	185.6	2	742.5	371
			16	2376	—	—	—	—	—	—	—	—	—	—	—	—
			18	2673	—	—	—	—	—	—	—	—	—	—	—	—
			24	3564	—	—	—	—	—	—	—	—	—	—	—	—
	240	297	8	2376	—	—	—	—	—	—	—	—	—	4	594	297
			10	2970	—	—	—	—	—	—	—	—	—	4	742.5	371
			16	4752	—	—	—	—	—	—	—	—	—	—	—	—
			18	5346	—	—	—	—	—	—	—	—	—	—	—	—
			24	7128	—	—	—	—	—	—	—	—	—	—	—	—

Resolution	Frame Rate	Pixel Clock	Color Depth	Total Data Rate	MachXO2, MachXO3L/LF, and MachXO3D (523 Mbps for RX and 728 Mbps for TX I/O)			ECP3 (467 Mbps for RX and 698 Mbps for TX I/O)			ECP5/ECP5-5G (467 Mbps for RX and 800 Mbps for TX I/O)			CrossLink (1.5 Gbps for RX and 1.5 Gbps for TX I/O)		
					Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock	Lane Number	Line Rate	Bit Clock
						Mbps	MHz		Mbps	MHz		Mbps	MHz		Mbps	MHz
FHD 1920x1080p (2200x1125)	60	148.5	8	1188	4	297	148.4	4	297	148.5	4	297	148.5	2	594	297
			10	1485	4	371	185.6	4	371	185.6	4	371	185.6	—	—	—
			16	2376	—	—	—	—	—	—	—	—	—	4	594	297
			18	2673	—	—	—	—	—	—	—	—	—	—	—	—
			24	3564	—	—	—	—	—	—	—	—	—	—	—	—
	120	297	8	2376	—	—	—	—	—	—	—	—	—	4	594	297
			10	2970	—	—	—	—	—	—	—	—	—	4	742.5	371
			16	4752	—	—	—	—	—	—	—	—	—	4	1188	594
			18	5346	—	—	—	—	—	—	—	—	—	—	—	—
			24	7128	—	—	—	—	—	—	—	—	—	—	—	—
UHD 3840x2160p (4400x2250)	30	297	8	2376	4	594 ¹	297	4	594 ¹	297	4	594 ¹	297	4	594	297
			10	2970	—	—	—	4	742.5 ²	371	4	742.5 ³	371	4	742.5	371
			16	4752	—	—	—	—	—	—	—	—	—	4	1188	594
			18	5346	—	—	—	—	—	—	—	—	—	—	—	—
			24	7128	—	—	—	—	—	—	—	—	—	—	—	—
	60	594	8	4752	—	—	—	—	—	—	—	—	—	4	1188	594
			10	5940	—	—	—	—	—	—	—	—	—	—	—	—
			16	9504	—	—	—	—	—	—	—	—	—	—	—	—
			18	10692	—	—	—	—	—	—	—	—	—	—	—	—
			24	14256	—	—	—	—	—	—	—	—	—	—	—	—

Notes:

- Meeting Tx MIPI Alliance specification, Rx is not meeting MIPI Alliance requirement, and Tsu/Thd window can be calculated based on the [tsu/thd Valid Window at Higher Data Rate](#) section. Rx side is 0.17 UI (0.287 ns * 594 Mbps) for MachXO2, MachXO3L/LF, and MachXO3D 0.19 UI (0.321 ns * 594 Mbps) for LatticeECP3 and ECP5/ECP-5G.
- Both Rx and Tx side not meeting MIPI Alliance specification, Rx Tsu/Thd window and Tx Tskew window can be calculated based on the [tsu/thd Valid Window at Higher Data Rate](#) and [Tskew Window at Higher Data Rate](#) sections. Tx side is 0.16 UI (0.215 ns * 742.5 Mbps), Rx side is 0.24 UI (0.321 ns * 742.5 Mbps).
- Meeting Tx MIPI Alliance specification, Rx is not meeting MIPI Alliance requirement, and Tsu/Thd window can be calculated based on [tsu/thd Valid Window at Higher Data Rate](#). Rx side is 0.24 UI (0.321 ns * 742.5 Mbps).

Reference

[MIPI D-PHY Interface IP \(RD1182\)](#)

Technical Support

For assistance, submit a technical support case at www.latticesemi.com/techsupport.

Revision History

Revision 1.2, May 2019

Section	Change Summary
—	Added Disclaimers section.
Device Selection	Updated Table 6.1. MIPI D-PHY RX/TX Implementation Hardware Comparison . - Specified Soft and hard implementations. - Added CrossLink (Hard) column. - Updated footnotes.
—	Changed instances of <i>MachXO3L</i> to <i>MachXO3L/LF</i> and <i>MachXO3D</i> .

Revision 1.1, March 2019

Section	Change Summary
All	Changed document title to MIPI D-PHY Bandwidth Matrix and Implementation.
Device Selection	Updated Table 6.1. MIPI D-PHY RX/TX Implementation Hardware Comparison . - Added IMIPI information for ECP5/ECP5-5G. - Updated information in the Number of Lanes row.
Back cover	Updated template.

Revision 1.0, November 2018

Section	Change Summary
All	Initial release. Note: Contents of this technical note were migrated from FPGA-UG-02041 (previously UG110). Publication of FPGA-UG-02041 user guide will be discontinued. Below are the changes from the previous FPGA-UG-02041 release (version 1.1). - Updated Table 6.1. MIPI D-PHY RX/TX Implementation Hardware Comparison. - Changed the RX Performance value for CrossLink. - Changed the TX Performance value for ECP5/ECP5-5G. - Corrected statement in MIPI Data Rate Calculation section to “The window for higher data rate does <i>not</i> meet MIPI Alliance Specification, but can still be practical in the user’s implementation.” - Corrected value in Figure 7.5. MachXO2 Maximum Data Rate to $T_{skew} = 0.661 - 0.455 = 0.206$. - Updated the ECP5/ECP5-5G section. - Corrected value from 820 to 800 in $Max\ Data\ Rate = 1/UI = 1/1.22 = \sim 800\ Mbps\ (at\ 0.15\ UI)$. - In Figure 7.7. ECP5/ECP5-5G Maximum Data Rate, changed $t_{SU_GDDR2_centered}$ to $t_{VB_GDDR2_centered}$ and $t_{HD_GDDR2_centered}$ to $t_{VA_GDDR2_centered}$. - Changed header value in Table 7.3. MIPI D-PHY Interface Lane Number and Line Rate Selection Example Matrix Table to CrossLink (900 Mbps for RX and 1250 Mbps for TX I/O)



www.latticesemi.com